



Society of Petroleum Engineers

SPE-229384-MS

Talent Transformation in Oil and Gas: Harnessing AI-Powered Digital Twins and Personas, Anchored in Change

Deepa Mary Satheesh, Weatherford International, Abu Dhabi, United Arab Emirates

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/229384-MS](https://doi.org/10.2118/229384-MS)

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

Abstract

Globalization and market volatility has warranted the critical need for managing talent and workforce in oil and gas industry. As the sector is cyclical agnostic, it requires business to maintain agility by closing the disconnect in digital transformation of its talent pool.

AI is transforming the way people work and while digital twins and personas have optimized asset performance management, predictive maintenance and remote monitoring, structured talent transformation remains in early stages. This paper strives to examine, explore and present a structured repeatable model for hybrid talent management strategy through AI -driven digital twins and AI-personas that can ensure a future-ready workforce aligned with business strategic priorities.

Drawing inspiration from adjacent industries and early adopters in energy sector – an architecture of deploying digital twins for high-risk operational roles (offshore rigs) and persona- based models for leadership and functional experts offers talent paradigm.

By leveraging HR analytics and AI-predictive modeling, organizations can create virtual simulations of high-risk and future workforce scenarios. A SWOT analysis, phased roll outs, cost-benefit analysis, structured change management can ensure seamless adoption of these technologies.

Adopting digital twins provides a virtual sandbox to simulate high-risk scenarios without real-life implications. Personalization is the new talent strategy as opposed to one- size - fit all approaches. AI personas shape leadership and strategic roles.

Supporting industry data states:

- 75% of large enterprises are investing in AI solutions (McKinsey, 2024)
- 85% of leaders emphasize personalization as key to talent competition (BCG, 2021)
- 20–50% reduction in training costs through AI-powered models (Deloitte, 2024)

The hybrid modeling unlocks significant finance and operational value by reducing costs 20% through improvements in talent pillars.

Digital twins are mirror replicas of employees while AI- personas are data-driven employee profiles. This paper introduces a novel combination to transform workforce development adoption through structured

transition from generic static programs to real-time, continuous adoption harness AI-based technology as catalyst for workforce agility. The model provides a scalable, measurable offering a new talent framework in the oil and gas sector while aiming to position the technology as enablers of enterprise talent pool.

Bridging emerging technologies from adjacent industries can pave the way for efficient, reliable and sustainable transformation of the workforce. The hybrid methodology -scalable in real time can reduce non-productive downtime, improve inefficiencies, close critical skill gaps and shape future learning pathways.

Early adoption will not only be a future -proof workforce but also accelerate organizations' competitive edge in global energy landscape.

Introduction

The energy industry is a powerhouse driving industrialization of the modern businesses, yet it faces cyclical challenges that demand improved operational optimization, workforce transformation, technological advancements in a dynamic global marketplace. With the proliferation and democratization of artificial intelligence (AI) across many industries, there has been a notable increase in profitability and productivity. This potential must be modeled and adapted specifically for the capital-intensive oil and gas sector.

According to [McKinsey & Company \(2023\)](#), AI has the potential to deliver 1.0 to 1.6% of productivity revenue in the energy sector, representing an estimated \$150-240 billion in additional revenue. [PwC's AI Jobs barometer report \(2025\)](#) highlights fourfold increase in productivity in AI-exposed industries and reports that 38% of energy professionals are or will be augmented by AI related technologies as part of their roles.

Industry 4.0 powered by AI and digital twins, call for reimagining business processes and employee experiences through a lens of future thinking - that will proactively shape the work experience while remaining resilient in dynamic energy landscape. Unlike other business sectors, AI does not disrupt the energy industry, rather can be leveraged granularly to reap benefits optimizing available resources strategically. The [Global Energy Talent Index \(GETI-2024\)](#) report exemplifies that specialists in the energy sector anticipate AI will transition their skills while requiring improvised digital literacy and critical thinking to meet evolving demands.

We are at the precipice of rapid digital transformation and that necessitates unparalleled digital literacy and talent adaptability. While advancements such as Digital twin technology, a spin-off advancement from the Internet of Things (IoT) are reshaping energy providers with streamlined performance of asset management and reduced costs for any unforeseen downtime, a critical gap persists in structured talent transformations.

This paper explores a conceptual framework intersecting digital transformation, evolution of workforce and organizational adaptation within the energy industry .AI will run the gamut of transformation by implementing a structured future focused talent framework enabling all stakeholders navigate the known knowns and emerging unknowns .Skill gap analysis will be foundation to determine workforce strategy .A dynamic approach within the foundation will equip talent upskilling ,reskilling and employee mobility .This transformation can then help with automation as well as augmentation of jobs .

Adopting AI-based talent frameworks delivers significant cost efficiencies, operational benefits and redefined talent paradigm thereby playing a pivotal role in navigating the paradoxical environment of energy sector, where conflicting forces and contradictory goals must be managed to successfully maneuver complex transitions.

A decade ago, the energy sector challenges revolved around cycles of busts and booms .Today, these issues are exacerbated by mounting wave of digital transition as companies strive to integrate the wave of digital fluency at all levels of the organization .While experienced workforce inherently offer deep operational expertise and institutional memory ,may face challenges in adapting to digital transformation initiatives -leading to slower growth in terms of digital adoption and training gaps .Conversely, younger ,

digital -native employees often lack industry-specific domain knowledge for maintaining operational excellence .As per Capgemini reports, 55% of global organizations acknowledge that the digital talent gap is widening. While the oil and gas sector has effectively navigated the competency gaps brought on by the "Big Crew Change" (Parshall, 2017), it now confronts a fresh set of challenges driven by digital transformation. This evolving landscape calls for a reimagining of talent strategies to ensure alignment with sustainable and future-focused industry practices. Against this backdrop, the following objectives can transform challenges into actionable strategies.

Objectives/Pillars

Build a skill-based organization

Transition from role based to skills-first structures to promote upskilling, reskilling, workforce agility and adaptability. For instance, automating skill gap analysis drives initiatives such as enhancing the digital literacy of employees, creating enablers, and preparing them for future challenges.

Establishing clear learning pathways

Fostering a culture of continuous learning through AI-curated and adaptive learning pathways ensures a resilient and responsive workforce capable of navigating business changes. By strengthening the organization's employee value proposition, AI-personas can personalize development journeys and engagement across diverse talent segments.

Digital talent marketplace and employee experience

By prioritizing transparent metrics, optimized job design, diverse talent pool, workplace safety and well-being, organizations can boost employee engagement and vitality. This contributes to improved retention and stronger human capital performance.

Strengthening leadership and development

Developing robust talent pipelines through targeted leadership programs. Equip high-potential talent to lead business transformations and position the organization as an industry pioneer.

Align Talent Practices with Strategic Organizational Goals

Ensuring HR, learning and talent initiatives are fully integrated with broader business objectives and operational excellence frameworks to sustain long term competitiveness. By identifying gaps and predictive analytics using AI, necessary changes can be mandated.

Embed ethics and data governance

As digital replicas revolutionize myriad industries, their growing influence necessitates embedding strong ethical and governance protocols for responsible AI use and trust in talent systems.

A Digital Twin Approach to Talent Management

While digital twins have revolutionized industrial operations like asset management within the oil and gas sector, their application to talent management offers a promising avenue for exploration. This proven approach can be capitalized on and extrapolated to talent management by adapting its core principles to human capital. Digital twins within talent space can be conceptualized as virtual replicas of HR processes, workforce groups and individual employees. As this is an emerging area within human capital management, this paper introduces the concept of digital twins - virtual replicas of talent workflows to bridge existing talent using AI. By understanding this parallel as illustrated in Fig 1, organizations can harness the power of digital twins to transform talent and build copious capabilities.

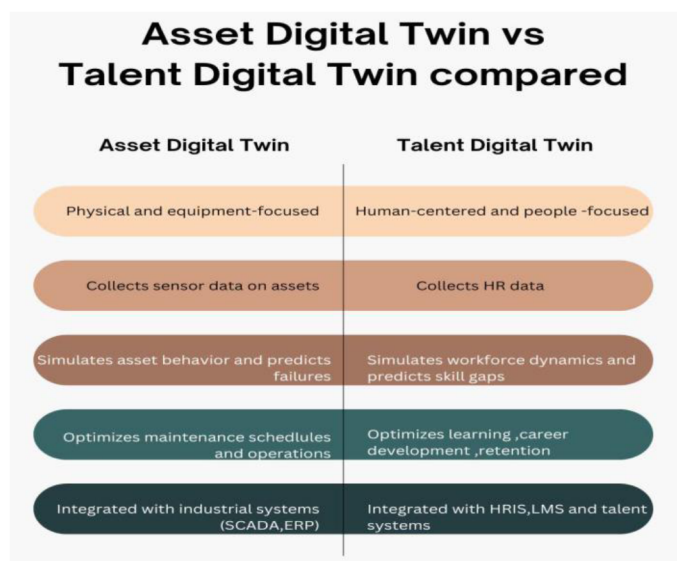


Figure 1—Applying the Digital Twin Concept to Talent Management

Hybrid Talent Management Framework

Leadership within the oil and gas companies is increasingly shifting from a directive style to catalyst model—one that emphasizes openness, transparency, and communication. This evolution enhances the Employee Value Proposition (EVP) and acts as a foundational precursor to more strategic talent management practice. A talent management framework proposed in this paper offers an empirically grounded roadmap. This serves as a vantage point to secure stakeholder buy-in and enable scalable, effective deployment by Talent teams within HR departments. Designed to be technology and infrastructure agnostic, this AI-enabled talent framework allows organizations to integrate seamlessly with existing systems and align with unique talent operating paradigms. The architecture outlines a flexible, scalable, and measurable approach to AI-enabled talent development.

Aligned with ongoing workforce digitization trends, the adoption of digital twins is accelerating and is illustrated as a three-tiered pyramid (see Fig 2). In this context, a digital twin acts as a simulation model that can analyze, predict, and optimize workforce decisions. It reflects a progressive deepening of digital twin adoption across HR maturity stages.

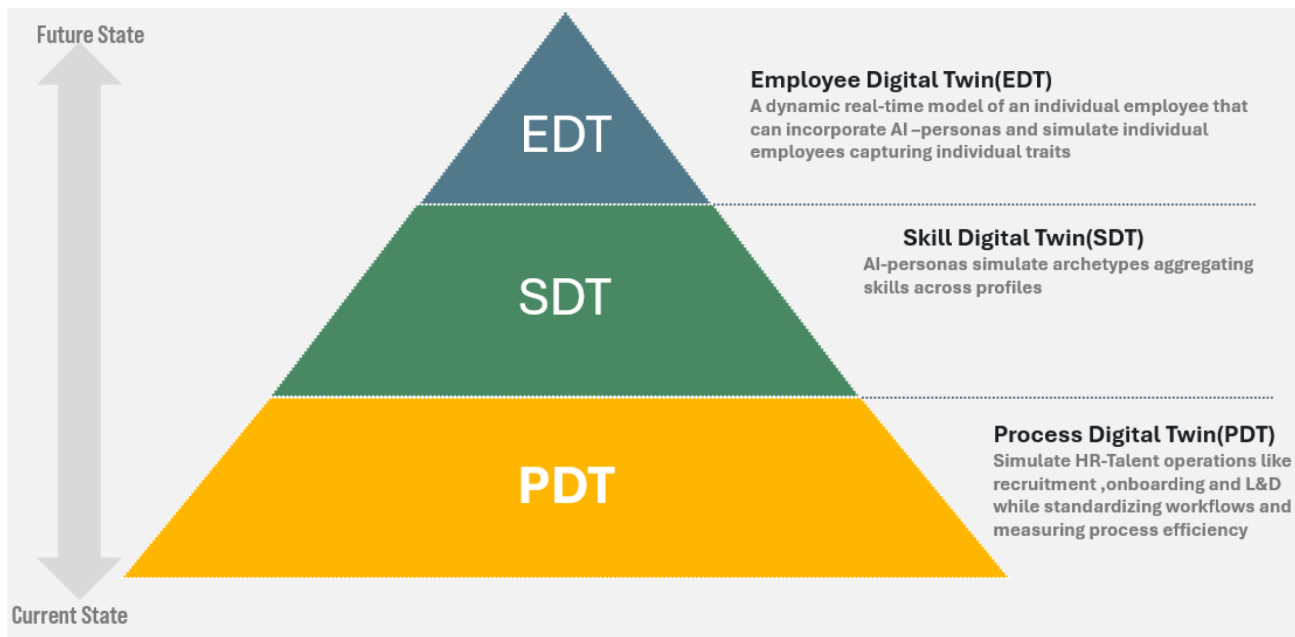


Figure 2—Digital Twin Evolution in Talent -From process simulation to individual cognitive

The holistic evolution of digital twins in HR or Talent enterprise can be classified as:

- **Process Digital Twins -PDT** at the base layer models organizational Talent processes such as recruitment, onboarding, learning and development and performance workflows (Deloitte, 2020). This layer simulates processes. To illustrate the practical impact, consider the case of Unilever that transformed its hiring process by leveraging AI-driven digital twin technologies .They achieved remarkable efficiency gains that translated to a 75-90% reduction in time to hire .It accelerated hiring cycles spanning several months to mere weeks in turn generating significant costs and workforce diversity setting a benchmark for digitally enabled talent management. (Forbes, 2025a)
- **Skill Digital Twins -SDT** at the middle layer advance toward modeling talent clusters based on transferable skills rather than static roles. As Hancock et al., 2022 note, many organizations are moving towards a skills-first approach in workforce design. This layer simulates group profiles. Boeing demonstrated a 40% increase in technician skill proficiency through immersive virtual training environments while reducing training time by 75% enabling workforce members to practice complex technical tasks effectively. (Forbes, 2025b)
- **Employee Digital Twins-EDT** at the top layer, also known as Digital Twin of an Employee (DTOE) represents the emergence of AI avatars created from individual behavior, cognitive and learning data (SHRM 2024). This apex layer is designed to simulate actual employees, personalize development, simulate trajectories and predict outcomes .AI-driven personas from SDT level will serve as a bridge enabling real-time personalization of employee experiences .It is important to note that this layer remains a future-forward concept with pilot experiments and studies in nascent stages .It is expected to evolve significantly as AI technology and organizational adoption mature .

The pyramid framework serves as an analytical lens for evaluating talent digitation practices across organizations, assessing adoption criteria at each layer. It uniquely integrates process, skills and employee-level models tailored for the energy sector.

Methodologies

In era of AI, organizational differentiation and competitive advantage within organizations will increasingly hinge on human capital workforce strategy, a key driver for value creation and sustain competitive advantage. As described by Fragiadakis et al. (2024), Human-AI Collaboration (HAIC) framework offers a strategic value to operationalize how organizations balance AI-driven insights from digital twins with essential human judgment. It helps define clear decision boundaries between automated intelligence and human expertise. Current digital twins are focused predominantly on managing technical assets only and seem to overlook the parallel need for talent strategies ensuring organizational adaptability.

As illustrated in Figure 2 (Deloitte, 2023), industrial digital twins enable strategic and operational decisions, scenario building and intelligent predictions.

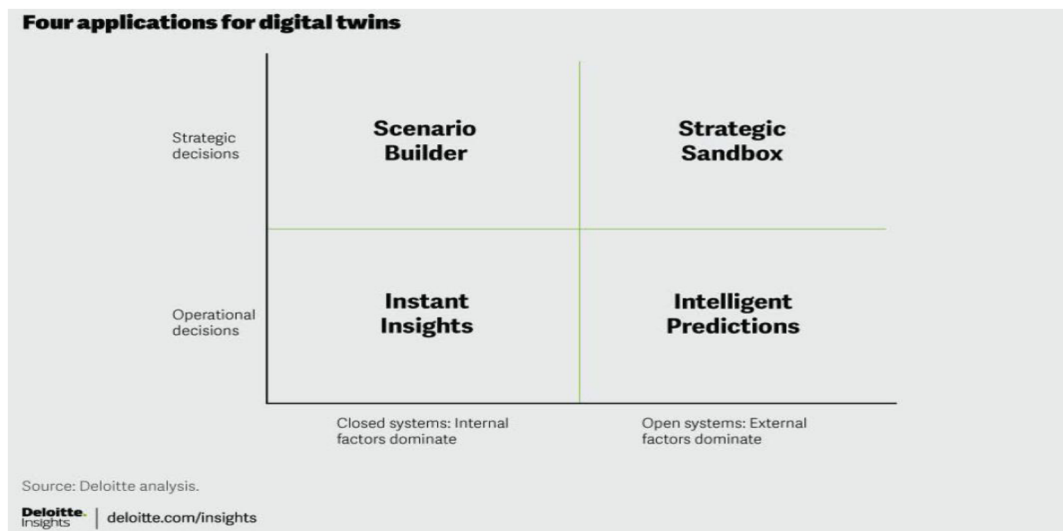


Figure 2—Digital Twin Simulation Matrix enabling decision making. (courtesy: New uses for digital twins | Deloitte Insights)

Building on the structural dimensions of Deloitte digital twin matrix, the below methodology is conceptualized to develop a talent strategy framework, focusing on parallel adoption of organizational talent. Companies can be myopic about their talent needs due to the brevity of change and complex milieu of digital acceleration, disruptive technology, energy transition and business dynamics. A digital twin of the talent process can help organizations in predicting potential bottlenecks of their workforce, help bridge gaps by analyzing and optimizing future workplace. Agile talent management efforts culminate in a cohesive approach that aligns with evolving uncertainties, organizational needs, and ongoing industry transformation – accelerating successful roll out of digital twin technologies.

Inspired by how artificial intelligence systems improve through repeated high-frequency exposure to diverse data sets, the paper underscores the necessity of designing a system where talent evolution is intentional and evidence based.

To stencil this, the proposed framework in Fig 3 is structured by four pillars:

- Talent Flow Optimiser
- Talent Ecosystem Simulator
- Future Workforce Blueprint
- Talent Market Intelligence

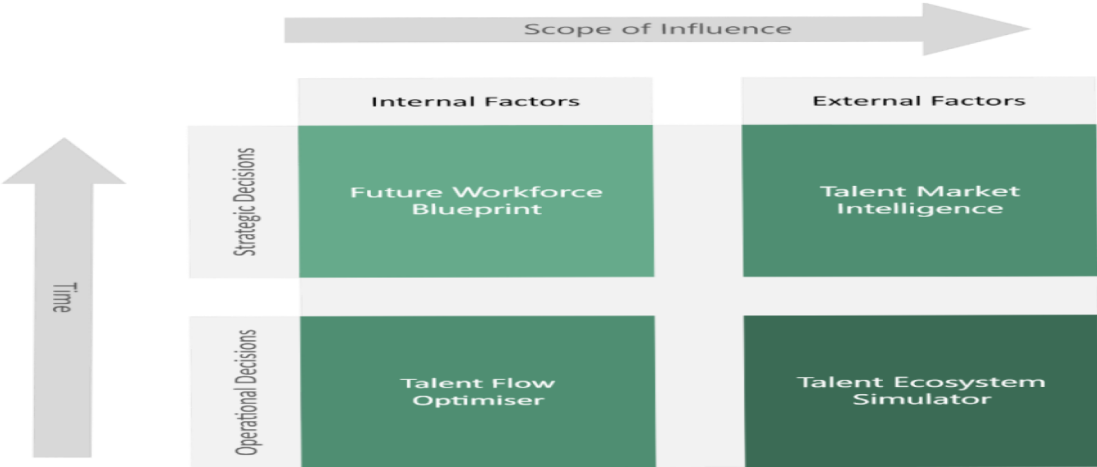


Figure 3—A talent strategy framework inspired by digital twin simulation models, illustrating the intersection of operational and strategic decisions influenced by internal and external dynamics of energy sector.

At the crossroads of operational decisions and internal factors, Talent Flow optimizers deliver real-time data on current performance metrics, serving as a mirror of internal talent workflows.

Progressing deeper into operational maturity is the Talent Ecosystem simulator, which integrates external factors-industry shifts, labor market trends-acting as precursors for strategic decisions. It represents a shift from reactive detection to proactive forecasting.

While the lower quadrants of framework align closely with existing enterprise systems for analytics, the upper quadrants are driven by digital twins offering a forward-looking blueprint of future workforce. These digital twins are powered by Talent Market Intelligence enabling businesses to anticipate and adapt to transformations.

Each quadrant of the talent matrix is amplified by a corresponding digital twin technology layer. This layered alignment allows for incremental buildup of capabilities and insights from automation and efficiency to long term strategic agility. Continuous feedback loops can be established. [Table 1](#) below highlights an integrated approach by explicitly citing each talent use case that will help organizations move from reactive to proactive HR forming resilient talent operating model, a metamorphosis of creating an ecosystem scaling from process to individual employee experience.

Table 1—Mapping talent areas across digital twin levels and decision quadrants

AI Talent Use Case	Digital Twin Level	Talent Flow Optimizer	Talent Ecosystem Simulator	Future Workforce Blueprint	Talent Market Intelligence
		Operational		Strategic	
		Internal scenario	External scenario	Internal scenario	External scenario
Workforce planning	PDT	Track resource allocation per team or project	Development with external partnerships	Simulate internal headcount and role needs	Model external labor availability by region
	SDT	Deploy teams based on skill mapping	Map functional roles with industry trends	Forecast role-based skill demands	Align group skills to external market trends.
	EDT	Forecast promotion readiness	Assess multi skill capabilities and fit in across industries	Predict individual mobility readiness	Simulate job transitions for key roles
Talent Acquisition	PDT	Automate recruitment process	Compare sourcing efficiencies vs competitors	Automate job requisition and approvals	Simulate sourcing timelines geographically
	SDT	Hiring vs upskilling	Workforce cost modeling	Group level hiring optimization and criteria	AI-based role fits from talent pools
	EDT	Automate offers of individuals, models employee readiness, career trajectory	Recommend gig roles	Predict best fit hires using AI personas	Simulate potential candidate career trajectory

Learning & Development	PDT	Automate training calendars, enrollment, feedback workflows. Deploy VR training module for technical sessions	Track outsourced training /vendor performance. Integrate simulation training with external partners. Academic institution collaboration	Align training plans with organizational objectives	Forecast market-driven learning demands
	SDT	Recommend courses for skill clusters such as AI sensitization. Experiential learning	Match cohorts with industry standard learning models	Build team-based learning paths. Forecast future roles requiring immersive skills and plan VR upskilling	Skills aligned with external certifications. Benchmark simulation training is used among peers.
	EDT	Nudges for learning adoption. Deliver personalized simulation-based learning paths for high -risk or technical roles.	Curate microlearning paths based on profile	Personalized learning journeys	Simulate future role alignment
Performance Management	PDT	Automate check-in reviews and feedback	Compare and implement frameworks with peers	Design future goal-setting frameworks	Benchmark KPIs with industry metrics
	SDT	Performance interventions by group	Apply external OKRs for internal teams	Calibrate team – level performance patterns	Mark success rates across teams
	EDT	Personalized feedback and recognition cycles	AI-generated peer benchmarking	Simulate potential contribution over time	Compare individual output to external benchmarks.
Succession & Mobility	PDT	Automate internal promotion process	Track external mobility options like exchanges	Simulate leadership pipeline and gaps	Compare leadership practices
	SDT	Recommend successors by role clusters	Dashboards for market readiness	Group potential	Cluster mobility trends
	EDT	High-potential tracking	Role match simulation for cross-border assignments	Individual succession simulation	Predict external leadership moves
Engagement & Retention	PDT	Automate pulse surveys, check-ins, stay interviews	Benchmark EVP components with external data	Engagement score modeling by organization layer	Compare retention practices globally
	SDT	Recommend group level retention plans	Apply cultural alignment simulations across locations	Team engagement trend modeling	Identify cultural drivers of team exits
	EDT	Tailored engagement journeys	Recommend external/internal career anchors	Personalized retention nudges	Predict individual exit risk vs opportunity
Offboarding	PDT	Automate offboarding & knowledge capture workflows	Track alumni engagement via external platforms	Model attrition impact on workforce flow	Benchmark exit processes across companies
	SDT	Standardize exit interviews as per role types	Alumni role matching with peers	Group-level attrition patterns	Predict talent movement patterns in industry
	EDT	Career redirection simulations	Link employee profiles to gig or rehire marketplaces	Personalized exit pathways	Simulate external placement options

Note: The PDT, SDT AND EDT applications shown for succession and mobility are conceptual use cases. Empirical solutions or pilot evidence do not currently exist and are highlighted to indicate directions for future research.

To effectively build and scale talent strategies across organizations, it is imperative to understand both the capabilities and specific use cases. By integrating industry talent management KPIs with AI-powered digital twin technologies - tailored for the oil and gas industry transformation journey - we can ensure relevance and impact. The strength of this approach lies in its empirical foundation, scalability and alignment with organizational priorities and emerging HR technology trends. [Table 2](#) highlights how data -driven adoption across the digital twin layers supports operational excellence and strategic agility in various talent management scenarios.

Table 2—KPIs and AI enablers for Integrated Digital Talent Framework

<i>TALENT DOMAIN</i>	<i>KPIs</i>	<i>Digital twin AI enablers</i>
<i>Workforce planning</i>	Resource utilization, headcount forecasting accuracy, skill gap closure	Predictive analytics, scenario simulation, labor market modeling
<i>Talent Acquisition</i>	Time to hire, quality of hire	AI driven talent matching, automated screening, career simulation
<i>Learning and Development</i>	Training adoption, effectiveness, skill proficiency growth	Personalized learning nuggets, VR modules
<i>Performance Management</i>	Goal achievement, peer benchmarking performance trend scores	AI feedback analysis, dynamic goal setting, outcome simulation
<i>Succession & Mobility</i>	Pipeline strength, promotion readiness, retention rate	Successor recommendation AI, high-potential tracking, mobility simulation
<i>Engagement & Retention</i>	Engagement levels, top talent retention, exit risk prediction	Sentiment analysis, survey analytics
<i>Offboarding</i>	Knowledge capture, alumni engagement	Career redirection modeling, knowledge automation, exit simulation

Change Management

Implementing new talent frameworks within the oil and gas sector requires careful change management to successfully align workforce capabilities with evolving business needs. Change management plays a vital role in preparing and supporting employees—regardless of their tenure or experience—with the skills, knowledge, and mindset needed to adapt to new processes and technologies that are otherwise opposed to skepticism. By integrating change management principles, organizations can enhance adoption rates, reduce resistance, and ensure that all employees become adept in applying the talent framework, thereby driving cultural transformation, operational excellence, and workforce agility.

Digital Employee Experience (DEX) encompasses the overall impression and perception employees form based on their interactions with the digital tools, system or platforms organizations provide to work efficiently. As enterprises increasingly adopt digital and AI solutions, optimizing DEX is critical to ensuring employee productivity, engagement, and satisfaction across all experience levels.

Successful adoption of AI-powered digital twins for talent transformation in oil and gas requires a thorough assessment of current organization capabilities, readiness and challenges ideally, the starting point. This dual assessment serves as the bedrock of change management while guiding decision making and tailoring implementation approaches.

The following aspects are considered crucial such as:

Leadership Engagement and sponsorship

Executive commitment is the foremost success factor for any transformation, ensuring alignment between business strategy and change outcomes.

Stakeholder engagement

Early and continuous involvement of all impacted stakeholders promotes ownership, reduces resistance, and supports seamless adoption.

Change readiness assessment

Conduct structured assessments to measure each talent use case for organization's preparedness and responsiveness, while minimizing fatigue from ongoing business transformations.

Communication Plan

Deploy timely, transparent and multi-channel communication strategies that deliver enhanced engagement and reinforce cause for change.

Training and capability building

Develop immersive learning experiences such as VR simulations, gallery walks to help ease expectations and management of digital transitions. For instance, evaluating the current learning and development of Next Generation Petro technical professionals and moving to experiential learning that thrives in AI-enabled digital ecosystems ([Agbaji, 2021](#))

Resource alignment and change champions

Assign designated change champions across functions to drive accountability, reduce ambiguity, and sustain momentum through every stage of transformation.

Cultural change initiatives

Shifting the mindset of workforce across generational cohorts is central to fostering openness to digital innovation. This requires more than just technical upgrades -it demands digital dexterity ([AlFedaghi, 2024](#)) where employees at all levels can be sensitized to transformations within.

Governance and risk management

Implement governance structures that address risk proactively ensuring compliance and trust within the change lifecycle.

Incorporating the Human AI collaboration in digital twin talent management helps organizations foster change by enabling them to adapt owing to inextricable symbiotic relationship between humans and AI. This change approach demarcates the boundary between machine insights and human expertise.

Recent research reinforces the value of champion roles within digital transformation. Industry reports like PwC's 2021 "Oil & Gas Digital Transformation" study found that oil and gas organizations which embed change champions and clarify ownership for each technology domain achieve faster digital twin adoption and improved upskilling outcomes.

Assigning champions for each Digital Twin Layer, as shown in [Fig 1](#) helps ensure sustained adoption across the digital twin continuum. These champions take ownership of the technology, people and change aspects tied to their assigned layer and across talent functions. As the organization's maturity progresses, these roles represent the human side of transformation, as demonstrated in [Table 3](#):

Introducing champions within existing HR -Talent to build familiarity across enterprise

Table 3—Functional Process Champions (Initial Roll out)

<i>Talent Process</i>	<i>Champion Role</i>	<i>AI-enabled Focus</i>
<i>Workforce planning</i>	Strategic HR Leads	Understand AI forecasting tools and basic analytics
<i>Talent Acquisition</i>	Talent Acquisition Leads	Support AI-driven resume screening and candidate matching
<i>Learning and Development</i>	L&D Partners	AI-leveraged recommendation of courses and track learning adoption
<i>Performance Management</i>	HRBPs	Facilitate AI-enabled feedback automation and trend spotting
<i>Succession & Mobility</i>	Talent Development leads	Use AI insights to highlight potential successors
<i>Engagement & Retention</i>	Employees experience lead	Utilize AI-driven sentiment analysis and attrition prediction
<i>Offboarding</i>	HR Managers	Exit data analysis and process automation using AI

As adoption deepens and systems mature, consolidate into focused champions per twin layer as illustrated below :

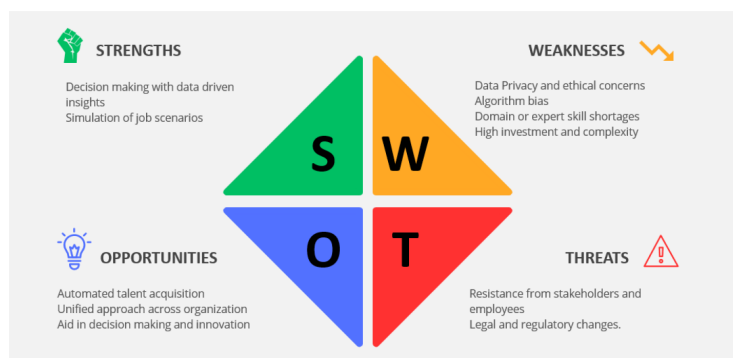
Table 4—Digital Twin Layer Champions-Post Stabilization

<i>Digital Twin</i>	<i>Champion Role</i>	<i>Focus</i>
<i>Process Digital Twin (PDT)</i>	HRIS Leads	System workflows, automation, and standardization
<i>Skills Digital Twin (SDT)</i>	L&D leads or skills framework owners	Skill aggregation, learning analytics, capability taxonomies, AI-personas
<i>Employee Digital Twin (EDT)</i>	HRBP or domain experts	Employee profiling, improved employee retention and accurate mobility predictions

Results

All energy companies are challenged with improved performance while treading business landscapes that are eclipsed by volatility, complexity, and ambiguity -VUCA (Rozman et al 2023). This catalyzes following a comprehensive talent -AI framework as a precursor in which workforce at organizations can incrementally adapt to evolving digital landscape fostering continuous learning. Business technologists and change champions can see skill enhancements while opening opportunities for talent upskilling and process improvements. The accelerated AI adoption can help dismantle traditional barriers and enable domain experts.

As per key observations illustrated through SWOT analysis in Fig 4, there is a fair share of identified weaknesses of implementing digital twins ranging from data privacy to complexities such as integration and user acceptance that need to be considered to initiate adoption not only as a single one-time initiative but one that is embedded in long term narrative.

**Figure 4—SWOT Analysis for Digital Twins in Talent Management**

A phased and measured approach supported by careful assessment of potential obstacles is essential to fostering sustainable adoption of digital twin technologies. Table 5 outlines key strategies to address challenges spanning technology, skills development, stakeholder engagement -critical factors for piloting a successful roll out. Clearly articulating the value proposition and linking to tangible business outcomes will help ensure the digital twin initiative remains robust, focused and aligned with organizational goals.

Table 5—Balanced opportunities/challenge analysis in digital twin implementation

<i>Weaknesses</i>	<i>Challenges</i>	<i>Opportunity for growth</i>
<i>Data privacy and ethical concerns</i>	Regulatory compliance, changing regional laws, data misuse, heightened cyberattacks	Establishing governance frameworks, ethical AI guidelines, enhanced security measures
<i>Algorithm bias</i>	Reinforce and perpetuate existing biases	Implementing bias detection, diverse data sourcing
<i>Domain or skill shortages</i>	Hampers adoption and innovation	Upskilling, partnership and outsourcing, talent pipelines
<i>High investment /Complexity</i>	Significant upfront costs, data quality, technical and operational integration hurdles with legacy systems, process shifts	Incremental pilots to manage risks, strategic collaborations, integrating cloud-based solutions for agility

Through strategic partnerships and collaboration ,the talent transformation can be executed diligently with both quantitative rigor and qualitative precision .Additionally, this also streamlines internal processing leading to operational excellence .Transparency extends to both internal and external stakeholders creating value and accountability .As organizations adapt to these measured changes it ensures bridging of talent loopholes to achieve industry leadership and carve a distinctive niche in the market.

Summary and conclusions

The overarching goal of the proposed framework establishes a compelling baseline for cross-functional initiatives that guide pilot implementations and phased deployments. With continued research and empirical validation, this work can serve as a springboard for further innovation, refinement of talent management practices, and successful large-scale implementation of AI-powered digital twin solutions across the oil and gas sector. Furthermore, by decoupling talent management framework from proprietary technologies, organizations can achieve future readiness but also gain a unique selling point through calculated technology overhauls, enabling them to lead talent transformation.

To succeed in an AI-driven future, organizations - particularly in the energy sector - must position themselves at the vanguard of transformation by leveraging AI-powered insights, investing in targeted upskilling and leading structured change initiatives. This pathway is critical to building digital dexterity among its workforce with essential knowledge, skills, and abilities to meet the industry demands. Business leaders need to go beyond deploying technology and proactively prepare their workforce for demands of the coming decade. By delving into the interconnected dimensions of technology, data and talent, organizations become privy to the headwinds that influence the market while driving competitive advantage and sustainable growth. Juxtaposing human intelligence with artificial intelligence requires deliberate strategies that build capability while fostering learning in support of innovation.

The convergence of AI, digital twins and predictive talent modeling signals the emergence of a Talent metaverse-an intelligent ecosystem designed to mirror, manage, and mobilize talent at scale across the industrial organization. While the ideal time to begin this transformation was a decade ago, the next best time is now. Establishing the digital backbone today is essential to securing a competitive edge in forthcoming years.

Acknowledgement

The author would like to thank Weatherford International for their support and permission to publish this paper.

References

1. McKinsey & Company (2023). The economic potential of generative AI: The next productivity frontier. Retrieved July 7, 2025 from <https://www.mckinsey.com/capabilities/mckinsey-digital/our-insights/the-economic-potential-of-generative-ai-the-next-productivity-frontier>
2. CPA2021 | WFPMA | World Federation of People Management Associations-BCG, 2021
3. PwC. (2025). Energy, utilities, and resources - PwC's 2025 global AI jobs barometer. Retrieved July 7, 2025 from *Energy, Utilities and Resources - PwC's 2025 Global AI Jobs Barometer*
4. Global Energy Talent Index (GETI) 2024: Future of Energy. Retrieved from <https://www.getireport.com/future-of-energy/>
5. Capgemini Research Institute. (2019). The digital talent gap: Are companies doing enough? *Capgemini*
6. Parshall, J. 2017. After Years, Big Crew Change' Has Passed, But Learning, Training Challenges Remain. *J Pet Technol* **69**: 38–40. <https://doi.org/10.2118/0717-0038-JPT>
7. The AI Recruitment Takeover: Redefining Hiring In The Digital Age, *Forbes* 2025a
8. Industry Leaders Are Employing Virtual Reality Training—Should You?, *Forbes* 2025b
9. Fragiadakis, G., Diou, C., Kousiouris, G., & Nikolaidou, M. (2024). Evaluating human-AI collaboration: A review and methodological framework. arXiv. <https://doi.org/10.48550/arXiv.2407.19098>
10. Deloitte Insights. (2020). Industry 4.0 and the digital twin technology: Smart factory applications and process analytic
11. Hancock, B., Higgins, C., Law, J., Olson, S., Patel, N., & Van Dusen, K. (2022, November 15). Taking a skills-based approach to building the future workforce.
12. Deloitte Insights. (2024). New uses for digital twins in the race to navigate an uncertain future-<https://www.deloitte.com/us/en/insights/topics/business-strategy-growth/racing-digital-twin-technology.html>
13. Society for Human Resource Management. (2024). Protecting digital personas. *SHRM. Protecting Digital Personas: A New Frontier in Employment Contracts*
14. Agbaji, Armstrong Lee. "Developing Next Generation Petrotechnical Professionals in the Age of AI." Paper presented at the Abu Dhabi International Petroleum Exhibition & Conference, Abu Dhabi, UAE, November 2021 <https://doi.org/10.2118/207813-MS>
15. AlFedaghi, Hadla S. "Digital Dexterity: Building Capabilities in the Era of AI." Paper presented at the ADIPEC, Abu Dhabi, UAE, November 2024 doi: <https://doi.org/10.2118/222277-MS>
16. PricewaterhouseCoopers. (2021). Oil & gas digital transformation-supercharging-digital-transformations-in-oil-gas.pdf
17. Artificial-Intelligence-Supported Reduction of Employees' Workload to Increase the Company's Performance in Today's VUCA Environment <https://doi.org/10.3390/su15065019>